



Mes-Allocs

A fintech solution designed to simplify benefit simulations.

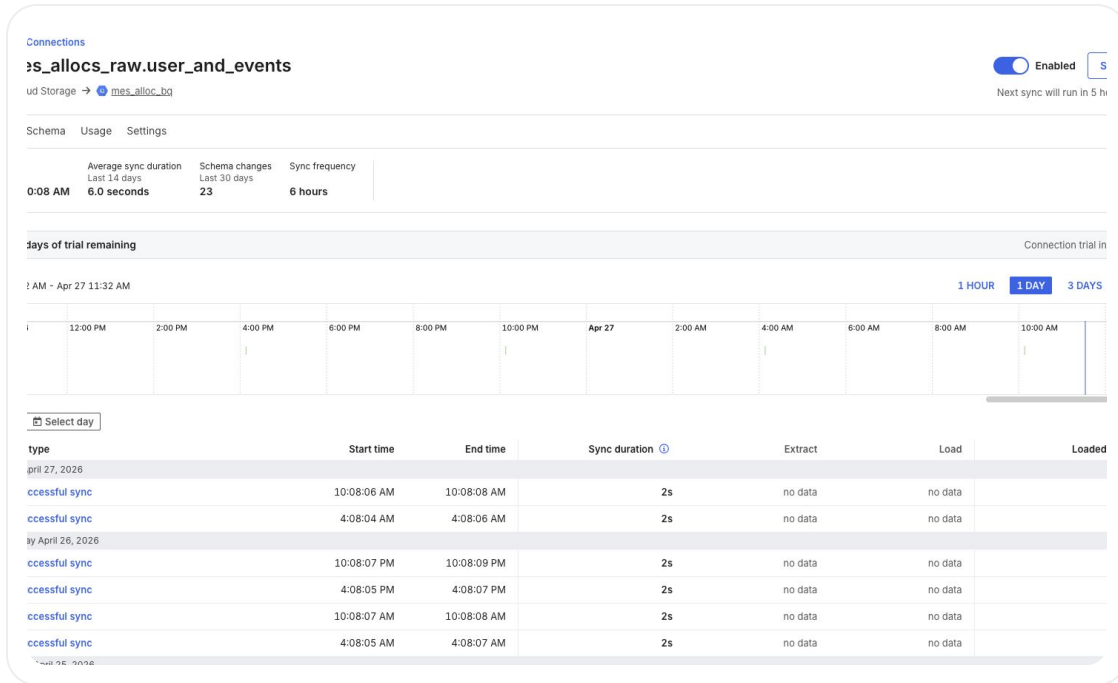
The Problem

"Users were dropping off the funnel, and we didn't know why."

Data Ingestion - The Foundation

I simulated **50k+ rows** of behavioral data using **Polars** to mimic a production environment.

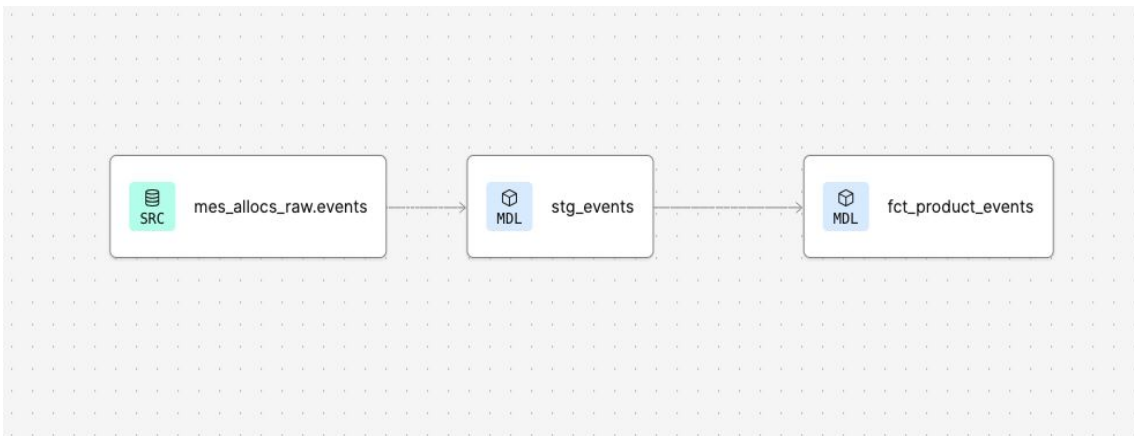
Connected **GCS to BigQuery** via Fivetran.



Dbt & Data Modeling - The “Brain”

Transformed raw logs into
"Gold" standard tables
using **dbt**.

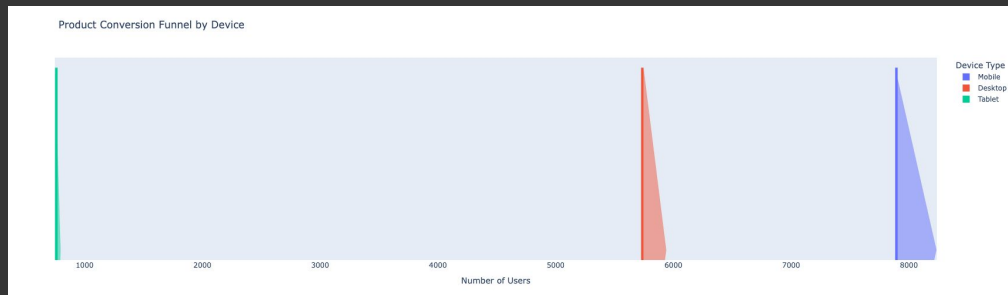
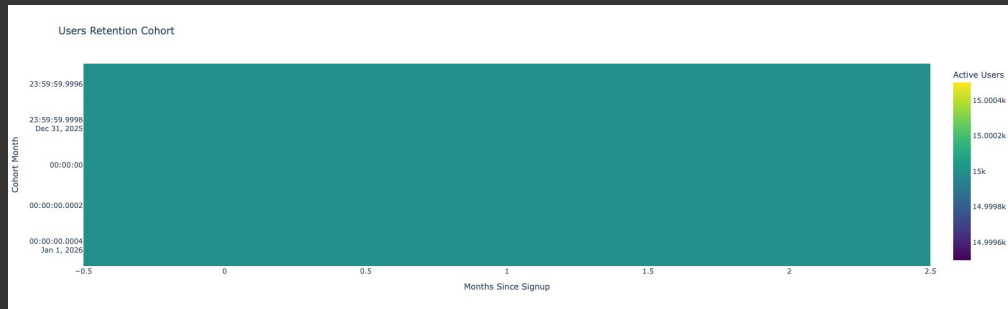
Implemented a **Medallion Architecture** (Staging → Marts).



Product Insights

Identified the "Leaky Bucket."

Discovered that **Mobile users** had a **15% lower** conversion rate than Desktop users.



The Prediction

Ran **A/B tests** to prove statistical significance.

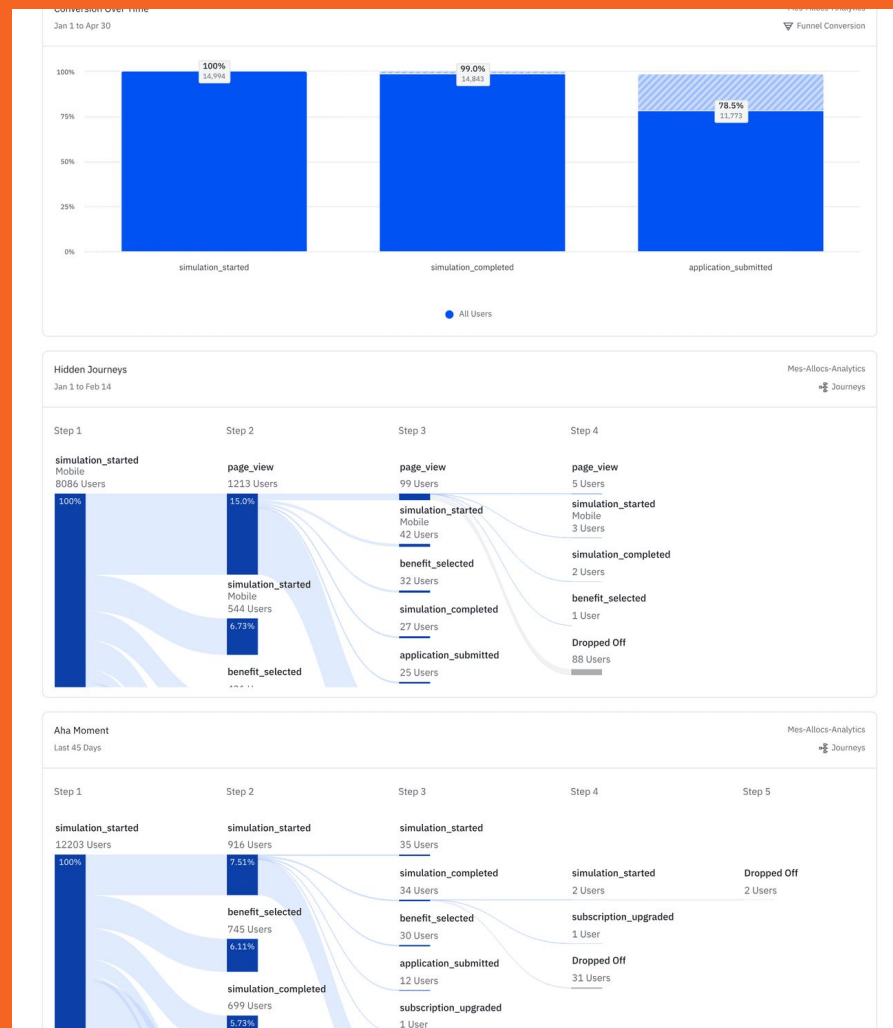
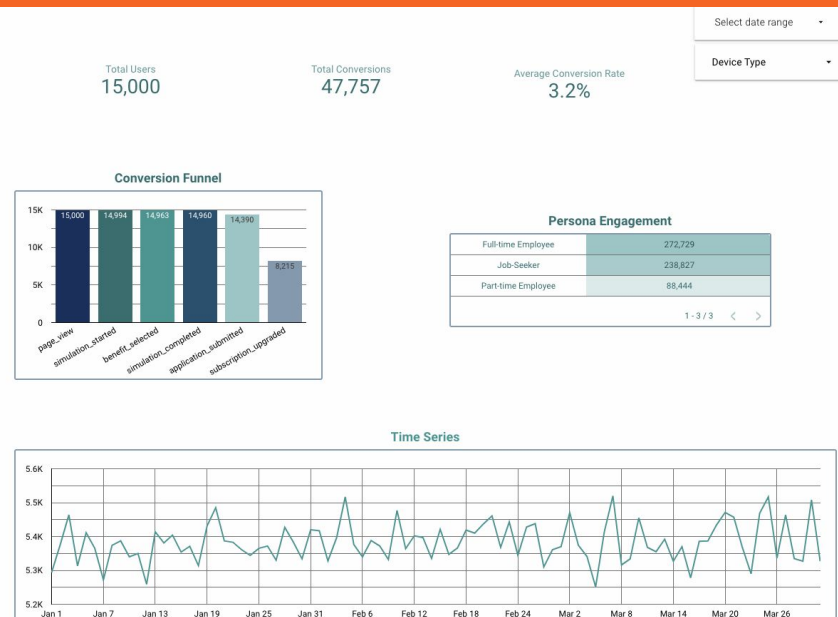
Built a **Random Forest Classifier** to predict user churn with **95% accuracy**.

	precision	recall	f1-score	support
0	0.29	0.01	0.03	141
1	0.95	1.00	0.98	2859
accuracy			0.95	3000
macro avg	0.62	0.51	0.50	3000
weighted avg	0.92	0.95	0.93	3000

ROC-AUC Score: 0.5610

The Visibility

Built a "Single Source of Truth" in **Looker Studio** for executives and used **Amplitude Pathfinder** to see exactly where users got confused.



The Action - Product Toolkit

User Search

Enter User ID



Mes-Allocs PM Command Center

Insights for user_0

Total Events

49

Status

Part-time E...

Plan

Basic



Predictive Churn Risk

Low Risk: 0.0% probability of churn

Recent Activity Timeline

	event_at	event_type	device_type
22	2026-03-30 07:20:05+00:00	page_view	Desktop
21	2026-03-29 09:34:58+00:00	page_view	Desktop
48	2026-03-27 15:37:32+00:00	simulation_started	Desktop
47	2026-03-26 08:20:45+00:00	simulation_started	Desktop
46	2026-03-21 15:52:20+00:00	simulation_started	Desktop

Final Impact & Recommendations

Based on the data, the recommendation is to **Redesign the Mobile Simulation Flow** and implement an **Automated Retention Discount** for users with a Churn Risk > 70%.

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